MEET BRICKBUDDY

Show the pile.
Pick a build. Change anything.

An AI building companion for children aged 6–8. Scan your bricks, pick a robot, then change it by voice or tap — and watch the instruction manual rewrite itself in real time.

THE BOOKLET

One-way paper.

- ✕ Can't see what bricks the kid actually has
- ✕ Can't answer "why?" or adapt mid-build
- ✕ Can't recolor when they want something different
- ✕ Only shows one "correct" model

→

BRICKBUDDY

A living manual.

- ✓ Scans the pile, recommends a build
- ✓ "Make it blue" — 3D and manual rewrite themselves
- ✓ Tap a brick to recolor only that one
- ✓ Teaches STEAM through every step

EVERY SESSION FOLLOWS ONE ARC · FIVE SCREENS, ZERO MENUS

📷

STAGE 1

Discover

Camera-first. Scan the pile, tell Buddy out loud, or just browse. One primary action, no text fields.

CAMERA · VOICE

🧱

STAGE 2

Inventory

An animated scan reveals every brick by color and count, then ranks three matched builds on what the kid actually owns.

SCAN · RECOMMEND

🔨

STAGE 3

Build ★

3D viewer + paper manual. Say *"make it blue"* or tap a brick — both 3D and manual rewrite themselves in real time.

3D · VOICE · TAP

🔬

STAGE 4

Learn

Five STEAM tabs grounded in what was just built. A two-question quiz lights up achievements, not grades.

SCI · TECH · ENG · ART · MATH

🏆

STAGE 5

Celebrate

Confetti, a rotating 3D trophy, a build report, and a printable certificate with the child's name on it.

REPORT · CERTIFICATE

SIX DESIGN DECISIONS — THE MOAT IS THE STACK

📖

Living manual

One mutation engine drives both 3D and printed manual. Voice, tap, structural — same source, never out of sync.

⚡

Local first

Every voice and tap mod is parsed deterministically on-device. Zero LLM stalls, zero offline failure.

🧩

Data bricks

Models are declarative JSON. The same 3D pipeline renders any robot the child imagines tomorrow.

🎯

Region aware

"Make the head purple" works because the parser knows aliases — head = snout = face. Voice scopes to a region; tap scopes to one brick.

🪶

Zero install

A URL, a webcam, a microphone. Works on a Chromebook, iPad, or any laptop. Deploys to Cloudflare's edge.

👶

Six-year UX

Big type, warm palette, always-visible mic, no hidden menus. Every screen is single-finger reachable.

Yusong Huang
FULL IMPLEMENTATION · MATERIALS
Engineered every screen — React, 3D viewer, AI pipeline, Cloudflare deployment — and produced this demo + poster + report.

Jiayi Huang
IDEATION · MATERIALS
Brought the original BrickBuddy concept, user research with children 6–8, and co-produced the demo

LIVE `brickbuddy.pages.dev`

REPO `github.com/Yusong-Enceladus/CMAA5043`

DEMO VIDEO `demo/video/BrickBuddy_investor_demo.mp4`

BUILT WITH React 19 · R3F · Three.js · Cloudflare Pages